

• linear SCM  $M = (X, E, W, P_E)$

- also known as linear additive noise model (ANM).

- structural equations are linear & represented using an upper triangular adjacency matrix

$$W \in \mathbb{R}^{d \times d} \rightarrow X = W^T X + \underline{E}(\text{noise})$$

•  $X$ : vector containing all  $d$  endogenous variables in the model

•  $W$ : Matrix where each entry  $W_{ij}$  represents the weight or strength of the causal effect from  $X_i$  to  $X_j$ .

•  $W^T X$ : direct causal influences, calculates value of each variable based on the values of the variables that cause it.

•  $E$ : vector of exogenous noise terms \* [when mapped through abstraction matrix  $T$ ]

• why we need (2) & (3) ↑? to prove  $M_H$  is a faithful T-abstraction of  $M_L$  authors must show  $M(e)$  &  $M_H(e)$  [reduced forms] are consistent.

• Formula 3

$$M(e) = F^T e \quad \text{where } F = (I - W)^{-1}$$

• formula 3 provides total effect view, allows you to skip the step-by-step interactions & calculate the final state of the system ( $X$ ) directly from the starting noise ( $e$ ).

-  $M(e)$ : reduced form of model, function that maps from specific noise config ( $e$ ) to the final observed values of all variables ( $X$ )

-  $F = (I - W)^{-1}$ : total causal effect matrix, unrolls all the internal loops & chains of command.

• If  $W$  tells  $A \rightarrow B$  &  $B \rightarrow C$ , then  $F$  will automatically show total effect of  $A$  on  $C$ .

-  $F^T e$ : linear transform. shows how a single spark of noise in one part of the system ripples through the entire network to produce the final results.

Formula (4):

-  $\tau: D(X) \rightarrow D(Y)$  [state mapping]

-  $\delta: D(E) \rightarrow D(U)$  [noise mapping]

for any low level interv.  $i \in I$  &

any exog. config.  $e \in D(E)$ :

$$\tau(L^i(e)) = H^{w(i)}(\delta(e)) \left. \vphantom{\begin{matrix} H^{w(i)}(\delta(e)) \\ \delta(e) \end{matrix}} \right\} \begin{matrix} \text{interventional} \\ \text{consistency} \\ \text{property} \end{matrix}$$

•  $w$ : intervention mapping

-  $w: I \rightarrow J$ , induced by the  
value abstraction func.  $\tau$

-  $w(i) = j$  iff  $Rst(H^j) = \tau(Rst(L^i))$

&  $j \in J$ , otherwise  $w(i)$  undefined

-  $\delta$ : exogenous abstraction function

•  $\tau(L(e)) = H(\delta(e))$  (5) - observational  
consistency equation.